

The background is a dynamic, comic-style illustration celebrating the 10th anniversary of League of Legends. It features a variety of champions in action. On the left, characters like Zed and Akali are prominent. In the center, a circular emblem marks the 10th anniversary. On the right, champions like Darius and Rengar are shown. The overall color palette is dominated by gold, blue, and grey, with a sense of movement and energy.

10
LEAGUE of
LEGENDS
YEARS

LEAGUE OF LEGENDS

A CRITICAL PSYCHOLOGICAL ANALYSIS

PRESENTER

ALFIE BUNYAN

MODULE

PSYM179

PROGRAMME

MSC PSYCHOLOGY IN GAME DESIGN

SCOPE AND SIGNIFICANCE

- ♦ **The MOBA Genre:** A complex intersection of real time strategy and role playing.
- ♦ **Cultural Impact:** 180 million monthly active players as of 2022.
- ♦ **The Ecosystem:** Beyond the game — esports, music, and the Arcane cinematic universe.

"How does a single game sustain 150 million players for over 15 years? The answer lies in the intersection of emergent design and weaponised motivation."

180M

MONTHLY PLAYERS

170+

CHAMPIONS

15+

YEARS ACTIVE

SECTION 02

CORE GAMEPLAY: INFORMATION PROCESSING

THE FIVE DECISION LAYERS

- ♦ Mechanical execution: last hitting, skill shots, positioning
- ♦ Champion knowledge: 170+ ability kits, cooldowns, scaling
- ♦ Map awareness: ward coverage, objective timers, enemy tracking
- ♦ Team composition: synergies, counters, champion select
- ♦ Meta reading: patch aware strategic choices and item builds

COGNITIVE LOAD THEORY (SWELLER, 1988)

EXPERT VS. NOVICE PROGRESSION

"Chunking reduces cognitive load as expertise develops (Miller, 1956). This staged mastery progression sustains long term engagement — the game never runs out of depth."

MILLER (1956) — CHUNKING AND WORKING MEMORY

EMERGENT COMPLEXITY

THE SIX CHAMPION CLASSES



ASSASSIN



FIGHTER



MAGE



MARKSMAN



SUPPORT



TANK

THE FIVE MAP POSITIONS



TOP



JUNGLE



MID



BOT



SUPPORT

THEORETICAL FRAMEWORK

Juul (2005): Simple rules generate emergent complexity. Designers cannot predict all strategies arising from systemic interactions.

MEANINGFUL PLAY

Salen and Zimmerman (2004): Meaningful play emerges from the interactions between designed elements, not from the elements themselves.



SECTION 03

RANKED SYSTEM: MOTIVATIONAL ARCHITECTURE

ELO RATING SYSTEM

LP gains and losses create variable outcome uncertainty. Hidden MMR sustains engagement loops.

ELO (1978)

SELF DETERMINATION

Autonomy, Competence, and Relatedness are all satisfied through the competitive climb.

RYAN & DECI (2000)

GOAL GRADIENT EFFECT

Motivation spikes as rank boundary approaches. Promo anxiety is a core design feature.

HULL (1932)

LOSS AVERSION

LP losses feel disproportionately painful, driving extended recovery sessions.

KAHNEMAN & TVERSKY (1979)

"Ranked does not just measure skill. It is a motivational architecture that exploits loss aversion and goal gradient effects simultaneously."

SKILL DEVELOPMENT AND FLOW

FLOW STATE (CSIKSZENTMIHALYI, 1990)

- ♦ LoL sustains flow via champion diversity: novices choose simple kits while experts choose high skill ceiling ones.
- ♦ Mastery ceiling remains perpetually distant: the flow channel is never permanently exited.

ATTRIBUTION AND LOCUS OF CONTROL

- ♦ Wins attributed to skill (internal) while losses attributed to teammates (external).
- ♦ Replay tools shift attribution internally: improving both performance and wellbeing.

"Internal locus: accepting responsibility correlates with faster rank improvement. 'Elo hell' as cognitive distortion preserves self esteem at the cost of growth."

PERSONAL TAKEAWAY

"Accepting internal attribution — that my losses were my responsibility — was the single biggest factor in my rank improvement."

TOXICITY: THE ARCHITECTURE OF AGGRESSION

DEINDIVIDUATION

Online anonymity and reduced accountability produce disinhibited aggression. Summoner name is not real identity.

ONLINE DISINHIBITION

Perceived invisibility amplifies antisocial expression. Text only communication removes vocal tone.

SOCIAL NORMS

Toxic majority establishes hostile communication as the community norm. New players are socialised into it.

FRUSTRATION AND AGGRESSION

LP loss and repeated failure trigger aggression toward the most proximate target: teammates.

"Toxicity in League of Legends is not a player character problem. It is a structural outcome of anonymity, high stakes, and loss aversion operating simultaneously."

INTERVENTIONS AND THEIR LIMITS

RIOT'S BEHAVIOURAL TOOLKIT

- ♦ **Honour system:** Skinner's (1953) operant conditioning for prosocial acts.
- ♦ **Instant Feedback:** ML based automated ban pipeline.
- ♦ **Restrictions:** Graduated sanctions for toxicity.
- ♦ **MUTE ALL:** Player controlled exposure reduction.

PSYCHOLOGICAL LIMITS

- ♦ **Learned helplessness:** Seligman (1975) repeated exposure leads to disengagement.
- ♦ **Punishment delay:** Delayed bans reduce the learning signal.
- ♦ **Attitude vs. behaviour:** Suppressing acts does not change attitudes.

CRITICAL REFLECTION

"Can design led systems alone solve a social norm problem, or does toxicity in LoL require community level cultural change that Riot cannot engineer from the top down?"

SECTION 08

MONETISATION: REINFORCEMENT REINFORCEMENT AND FOMO

VARIABLE RATIO

Hextech Chests use gambling psychology. Unpredictability creates extinction resistant reinforcement.

SKINNER (1953)

PRESENT BIAS / FOMO

Limited time events engineer urgency spend, forcing immediate purchase decisions.

O'DONOGHUE & RABIN (1999)

OVERJUSTIFICATION

External rewards erode intrinsic motivation. The game is played for the reward, not the experience.

LEPPER ET AL. (1973)

ARTIFICIAL SCARCITY

Legacy Vaults and Prestige retirements sustain social comparison pressure.

CIALDINI (1984)

"A system where players describe spending to avoid missing out, not because they actively wanted the item. The want is manufactured."



THE ETHICAL PROBLEM IS NOT ABSTRACT

"67% of League of Legends player base aged 16 to 24: the population most neurologically vulnerable to variable ratio reinforcement."

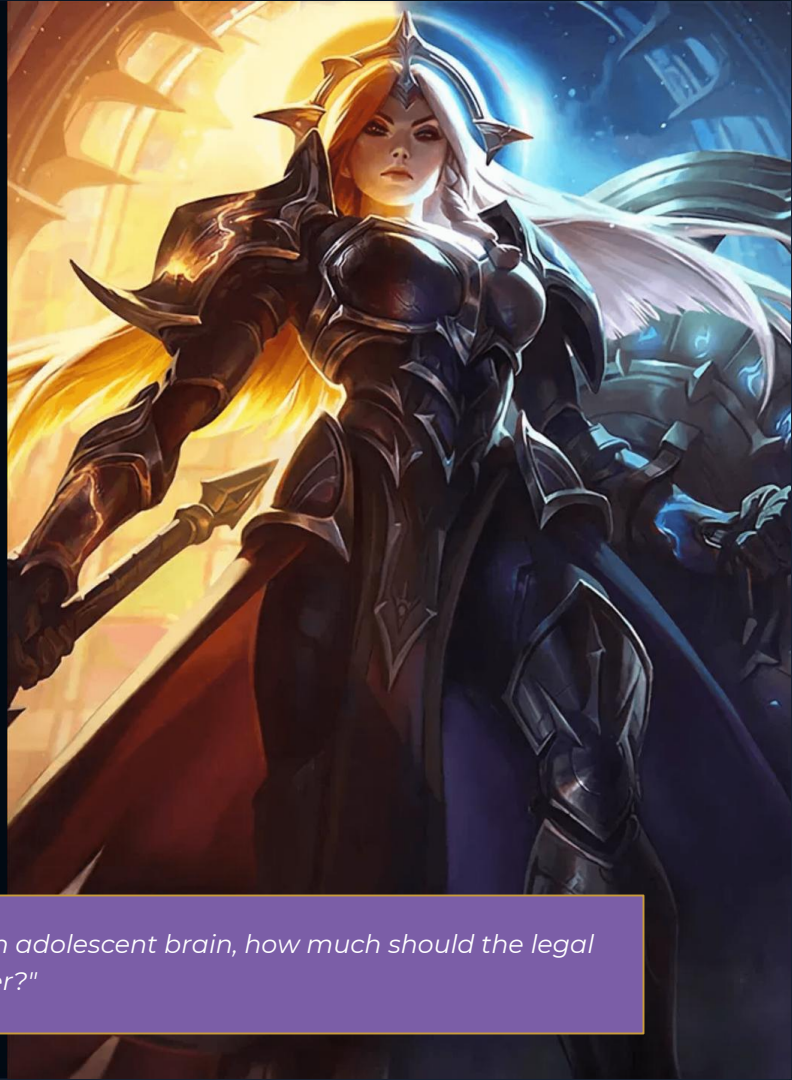
ADOLESCENT BRAIN DEVELOPMENT

- ♦ Prefrontal cortex not fully developed until mid 20s: elevated impulsivity.
- ♦ Heightened reward sensitivity: variable rewards proportionally more potent.

REGULATORY POSITIONS

- ♦ **Belgium (2018):** Classified loot boxes as gambling; Hextech mechanic banned.
- ♦ **UK (2017):** Not classified as gambling; no cash out mechanism.

"If a system produces the same behavioural responses as gambling in an adolescent brain, how much should the legal definition of gambling matter?"



TWO SYSTEMS, ONE ECOSYSTEM

01 ♦ EMERGENT DESIGN: PLAYERS AS CO-DESIGNERS

Riot provided the architecture while 150 million players built the culture. Bi-weekly patches manage emergent behaviour rather than prescribe it. This is a designed system for perpetual novelty.

02 ♦ RANKED PSYCHOLOGY: WEAPONISED MOTIVATION

Ranked satisfies all three SDT needs while simultaneously weaponising loss aversion and goal gradient effects. Toxicity is a structural output of this motivational system.

03 ♦ MONETISATION: CONDITIONING AND ETHICS

Hextech loot boxes deploy variable ratio reinforcement and FOMO mechanics against a player base whose adolescent neurology makes them disproportionately vulnerable.

"The players built the meta. Riot built the model."

SECTION 10

REFERENCES

◆ CORE THEORY

Berkowitz, L. (1989). Frustration aggression hypothesis: Examination and reformulation. *Psychological Bulletin*.

Casey, B. J., et al. (2008). The adolescent brain. *Annals of the New York Academy of Sciences*.

Csikszentmihalyi, M. (1990). *Flow: The Psychology of Optimal Experience*. Harper & Row.

Hull, C. L. (1932). The goal gradient hypothesis and maze learning. *Psychological Review*.

Juul, J. (2005). *Half Real: Video Games between Real Rules and Fictional Worlds*. MIT Press.

Kahneman, D., & Tversky, A. (1979). Prospect Theory: An Analysis of Decision under Risk. *Econometrica*.

Lepper, M. R., et al. (1973). Undermining children's intrinsic interest with extrinsic reward. *Journal of Personality and Social Psychology*.

◆ DESIGN & MOTIVATION

Elo, A. E. (1978). *The Rating of Chessplayers, Past and Present*. Arco.

Postmes, T., & Spears, R. (1998). Deindividuation and antinormative behavior: A meta analysis. *Psychological Bulletin*.

Riot Games (2009). *League of Legends* [Video Game].

Rotter, J. B. (1966). Generalized expectancies for internal versus external control of reinforcement. *Psychological Monographs*.

Ryan, R. M., & Deci, E. L. (2000). Self determination theory and the facilitation of intrinsic motivation. *American Psychologist*.

Salen, K., & Zimmerman, E. (2004). *Rules of Play: Game Design Fundamentals*. MIT Press.

Seligman, M. E. P. (1975). *Helplessness: On Depression, Development, and Death*. Freeman.

Skinner, B. F. (1953). *Science and Human Behavior*. Macmillan.

◆ ETHICS & REGULATION

Suler, J. (2004). The Online Disinhibition Effect. *CyberPsychology & Behavior*.

Sweller, J. (1988). Cognitive load during problem solving. *Cognitive Science*.

Weiner, B. (1985). An attributional theory of achievement motivation and emotion. *Psychological Review*.

Zimbardo, P. C. (1969). The human choice: Individuation, reason, and order versus deindividuation, impulse, and chaos. *Nebraska Symposium on Motivation*.

Belgian Gaming Commission (2018). *Research Report on Loot Boxes*.

UK Gambling Commission (2017). *Virtual currencies, eSports and social casino gaming*.

Zendle, D., & Cairns, P. (2018). Video game loot boxes are linked to problem gambling. *PLOS ONE*.